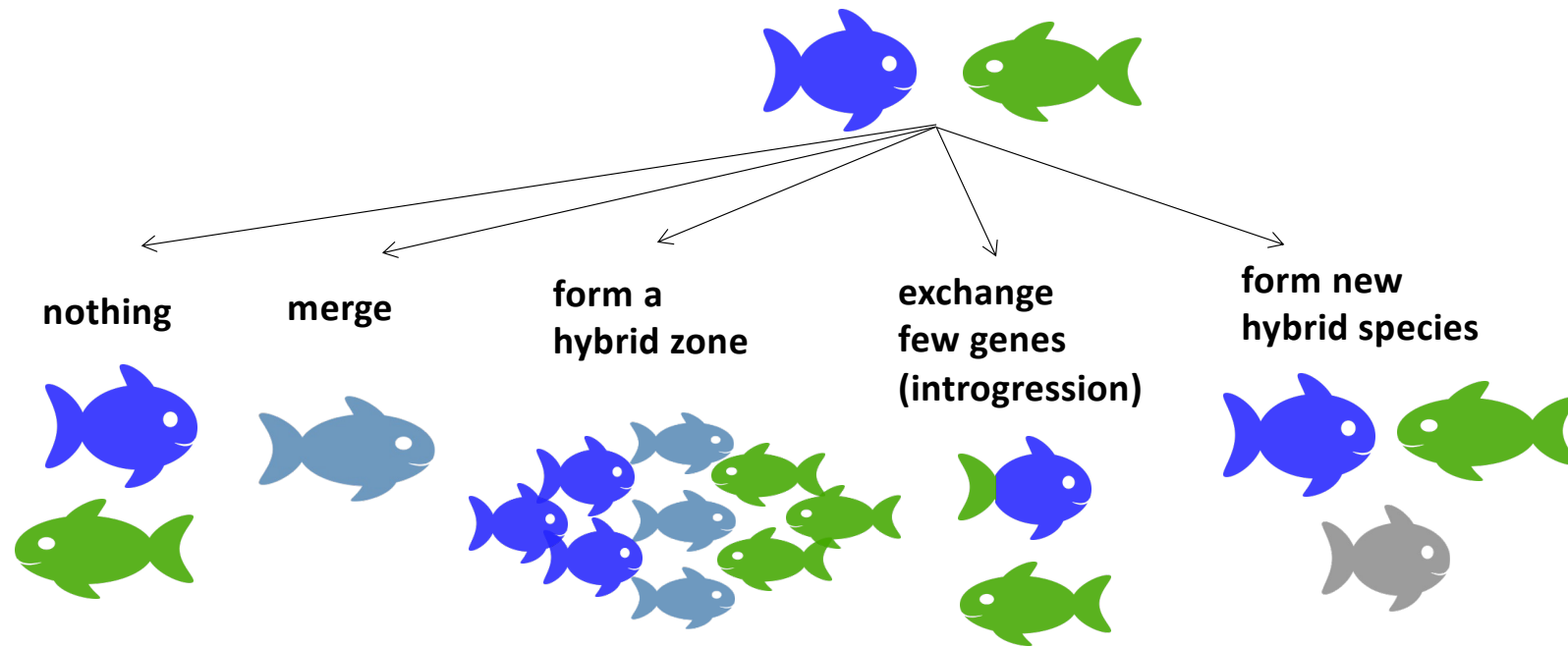


Inferring hybridization (and introgression)

Joana Meier & Fernando Seixas

Evolutionary consequences of hybridisation



Hybridisation at different time and divergence scales

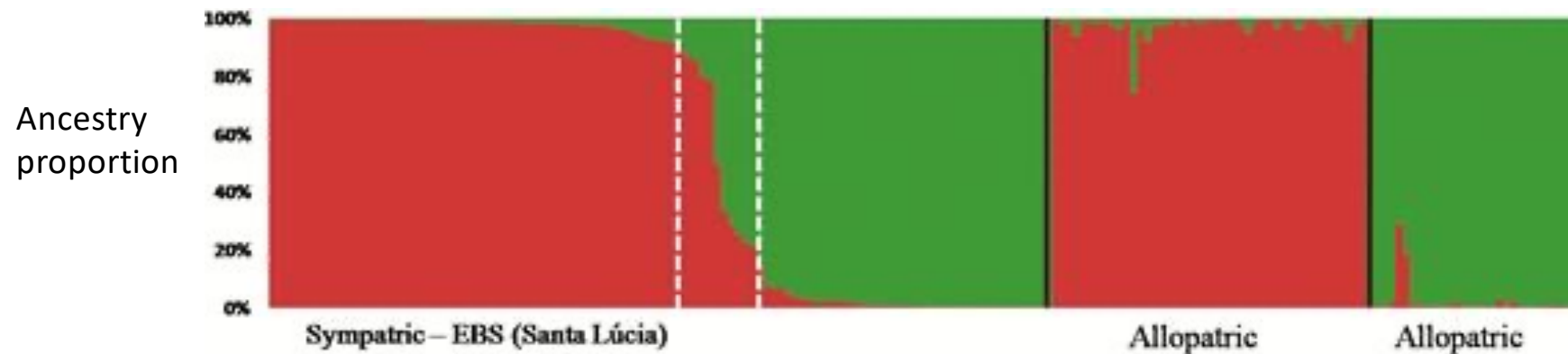
- **Current ongoing hybridisation:**
 - **PCA** to get a first overview
 - **STRUCTURE/ADMIXTURE/fastSTRUCTURE** to get a general overview and infer admixture proportions for each individual
 - **Triangle plots** to classify hybrids
 - Potentially **cline analyses** if the hybrids are geographically restricted

Example of clustering methods to identify hybrids and infer ancestry proportions: e.g. bromeliads

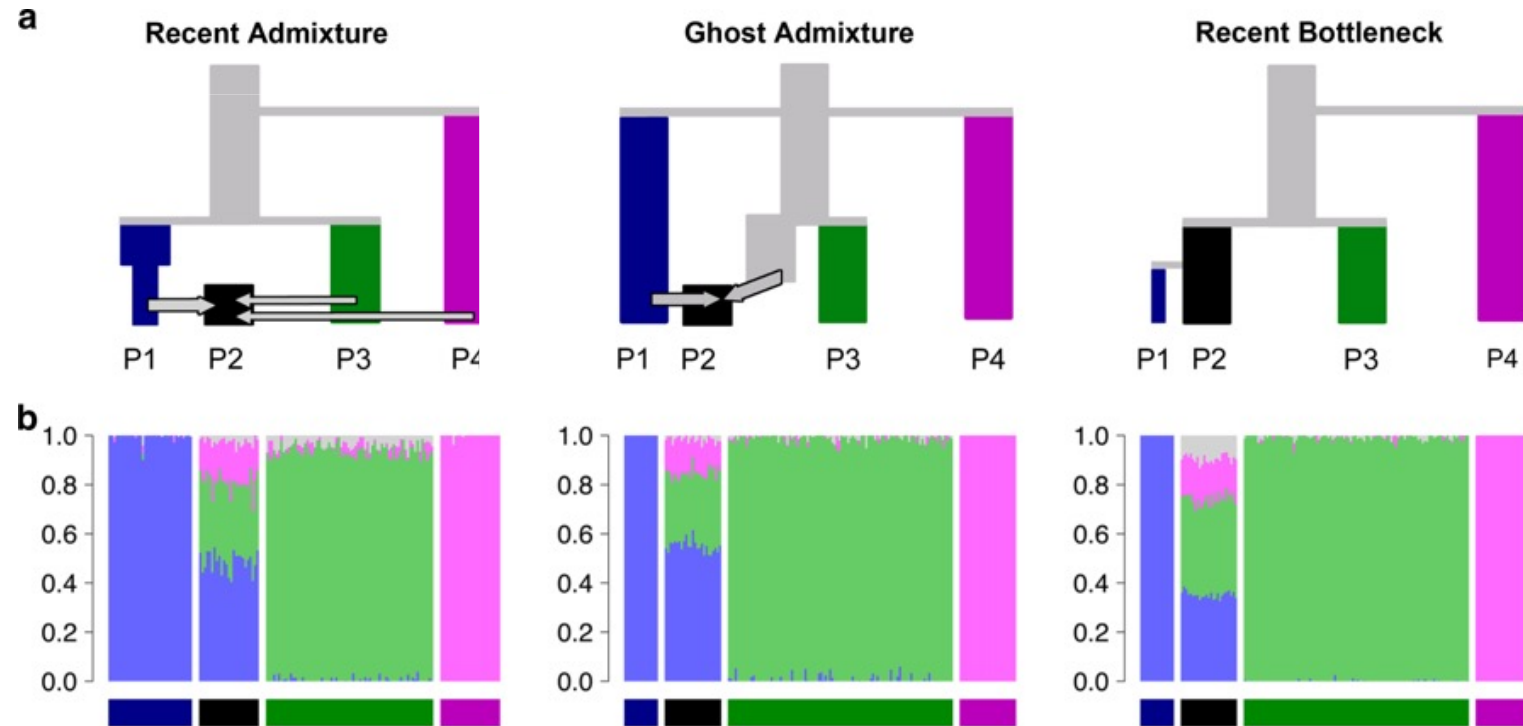


Vriesea simplex

Vriesea scalaris

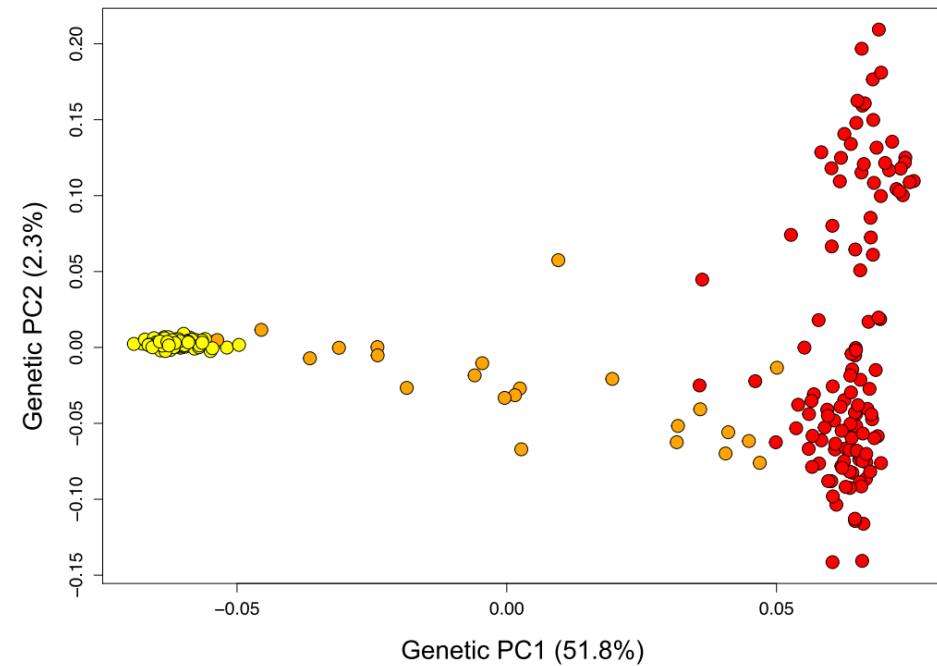
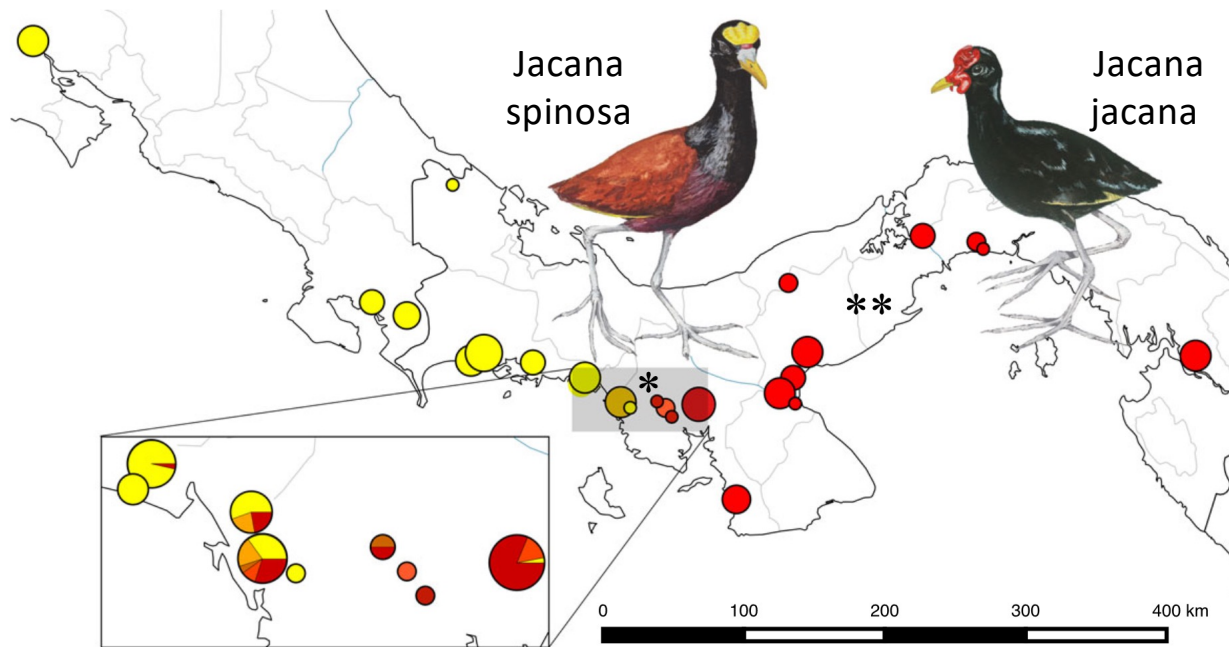


Very different evolutionary histories can give the same ADMIXTURE/STRUCTURE plot



Very important: Use STRUCTURE/ADMIXTURE plots only to get a first overview of genetic structure. Putative cases of hybridization need to be followed up with other analyses such as D statistics and/or demographic modeling. Potentially use 'badMixture' (by Dan Lawson) to check how well your result fits the data.

Triangle plots help to identify recent hybridisation (e.g. triangulaR)



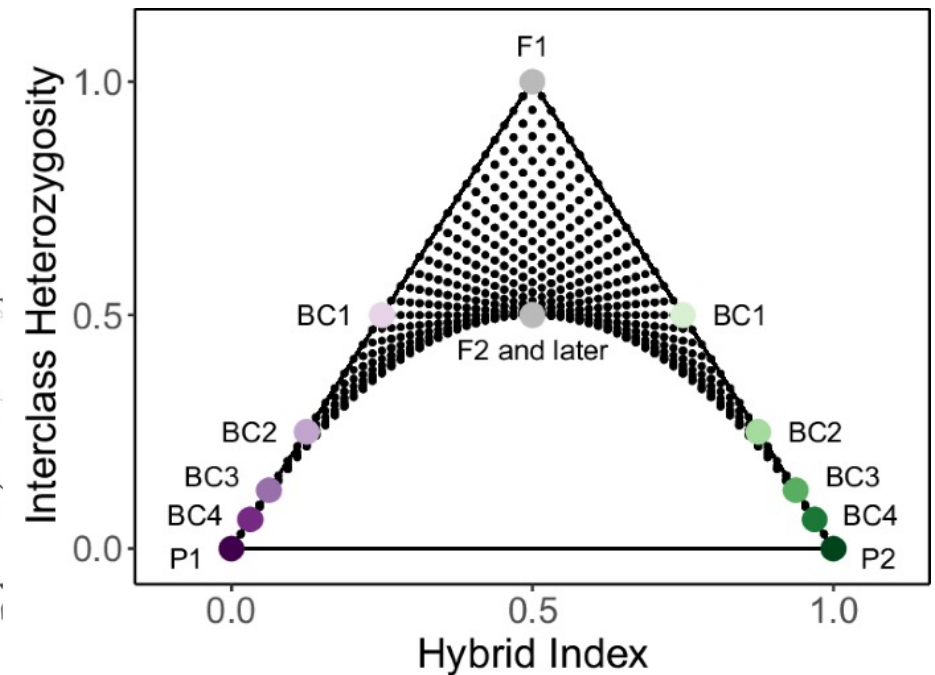
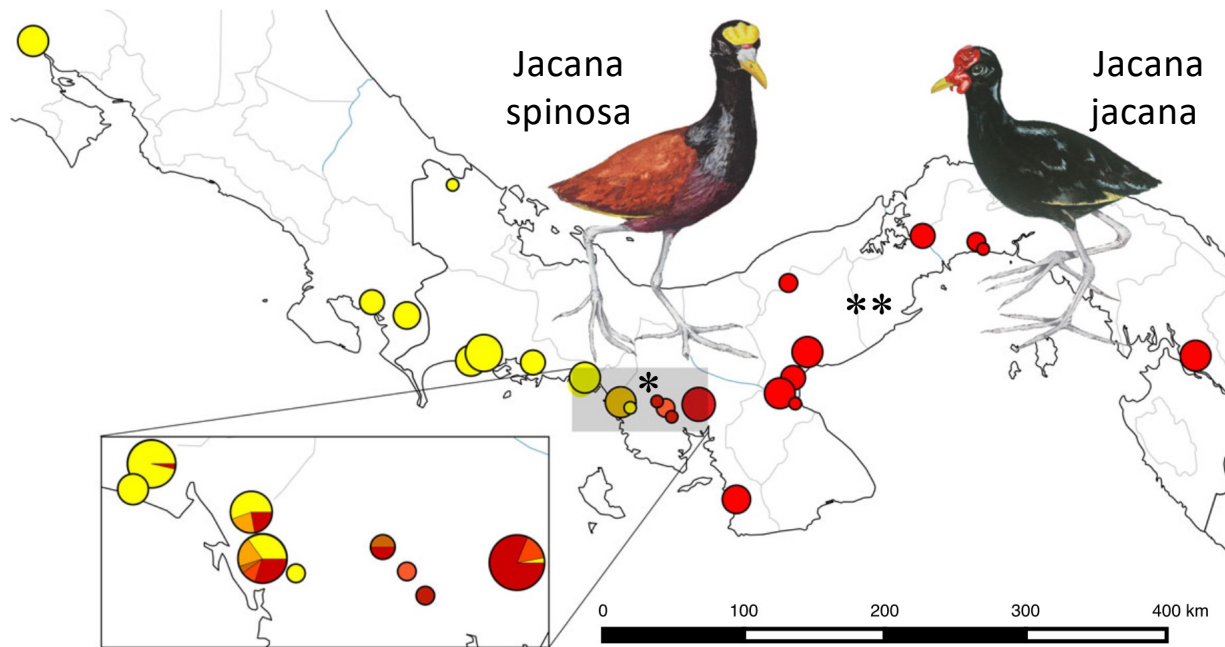
Alternatives to triangulaR:

mongrail2: <https://github.com/mongrail/mongrail2>

Hlest: <https://rdr.io/cran/Hlest/>

Lipshutz et al. 2019

Triangle plots help to identify recent hybridisation (e.g. triangulaR)



ancestry-informative markers (AIM)

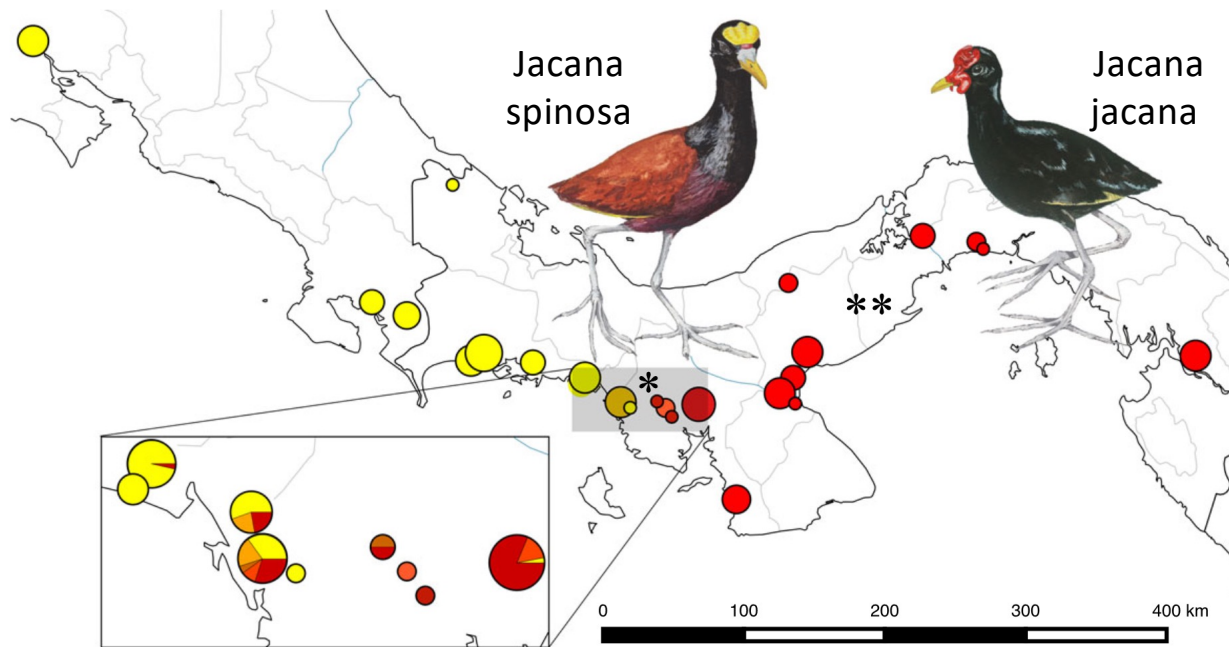
Alternatives to triangulaR:

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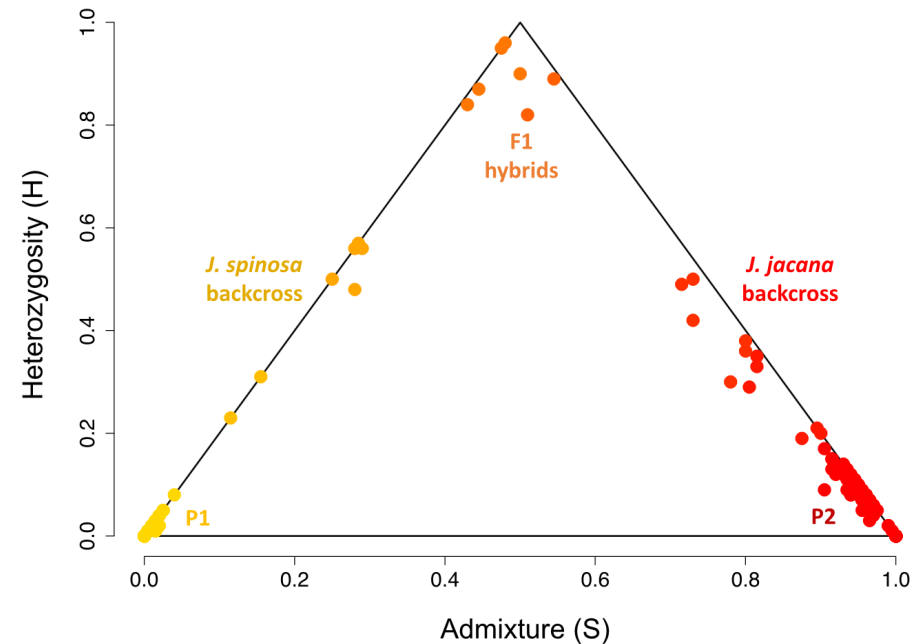
Hlest: <https://rdr.io/cran/Hlest/>

Lipshutz et al. 2019

Triangle plots help to identify recent hybridisation (e.g. triangulaR)



Triangle plot shows asymmetric introgression



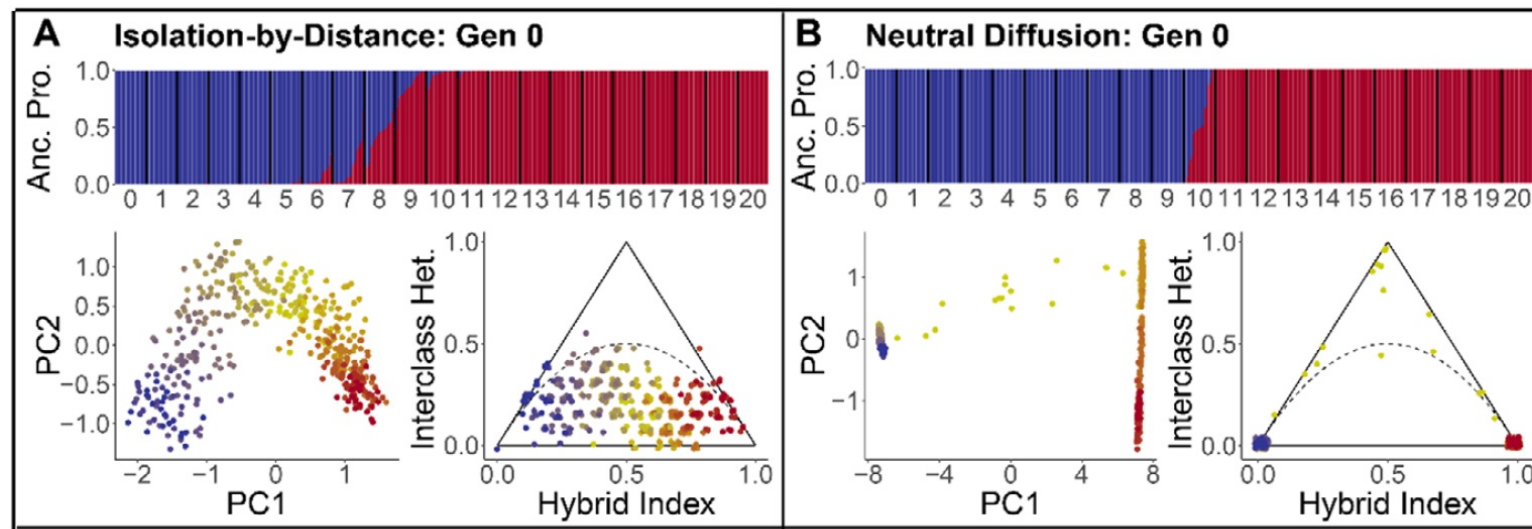
Alternatives to triangulaR:

mongrail2: <https://github.com/mongrail/mongrail2>

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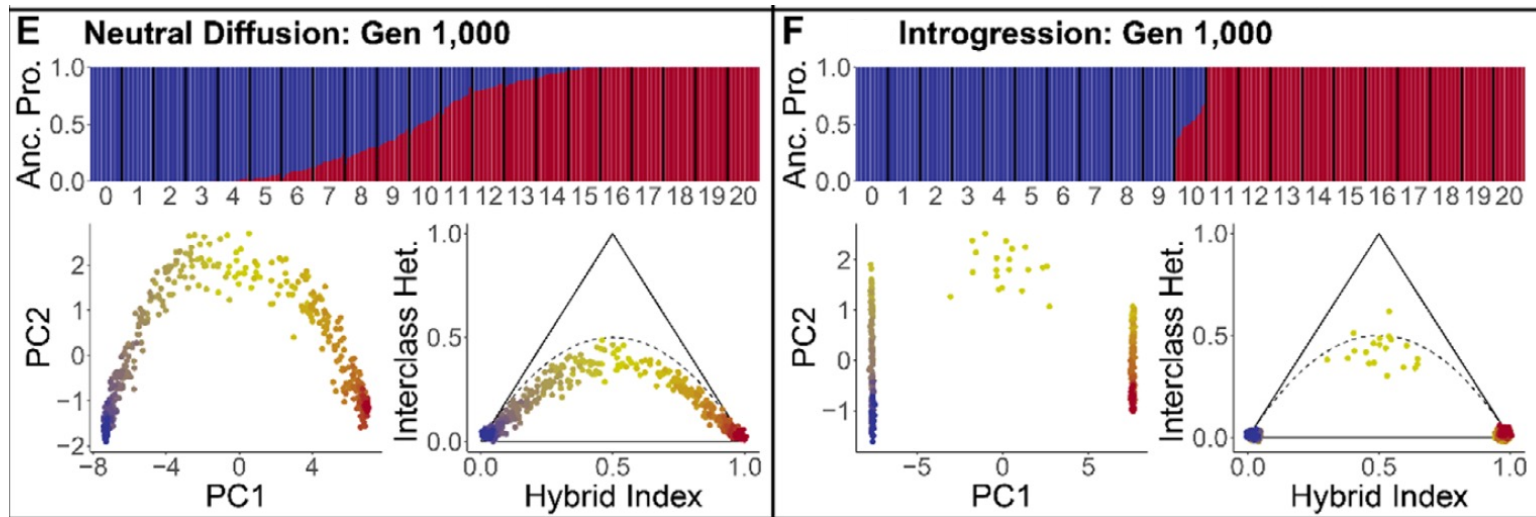
Lipshutz et al. 2019

Also strong isolation by distance can mimic hybridisation in STRUCTURE plots -> use triangle plots (e.g. triangulaR)

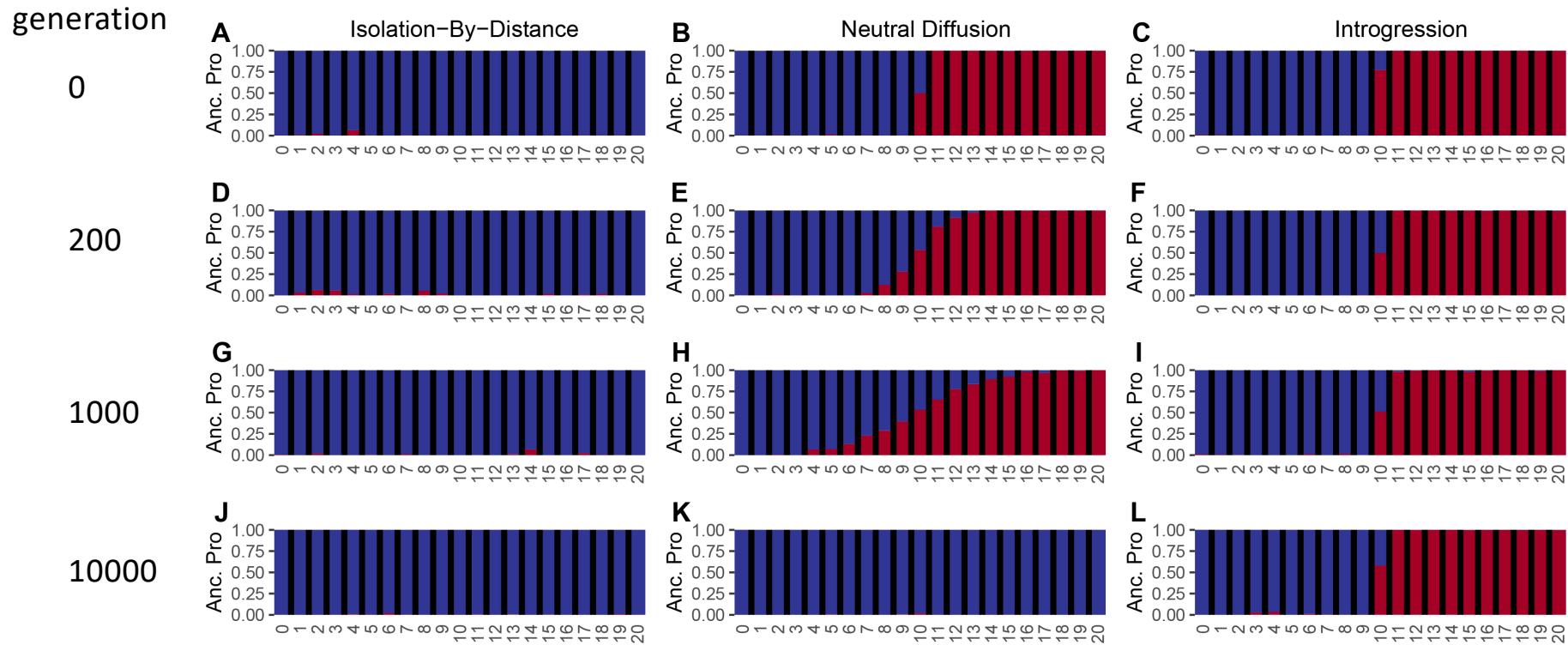


Isolation by distance can mimic hybridisation in STRUCTURE plots -> use triangle plots (e.g. triangularR)

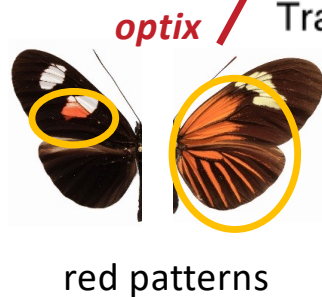
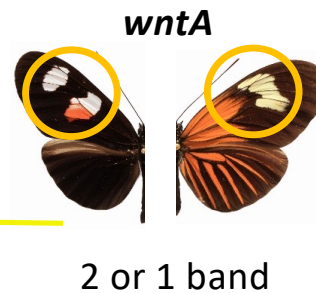
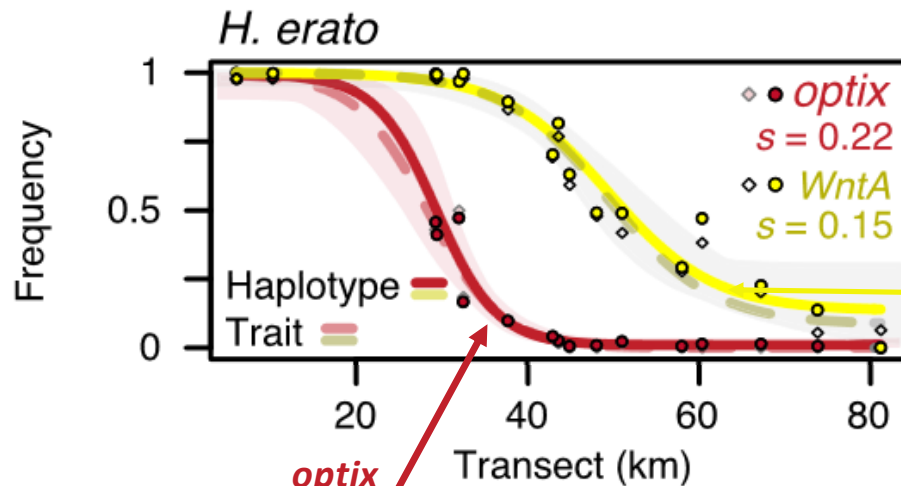
1000 generations after the initial contact: we get all kinds of hybrids if there is no reproductive isolation or hybrids restricted to a small hybrid zone if there is reproductive isolation.



conStruct (Bradburd, Coop, and Ralph 2018) allows distinguishing isolation by distance from introgression



Hybrid zones allow us to test for barriers to reproductive isolation and estimate selection



$$\sigma \sim w \sqrt{s/12}$$

Dispersal rate

Cline width

Selection coefficient

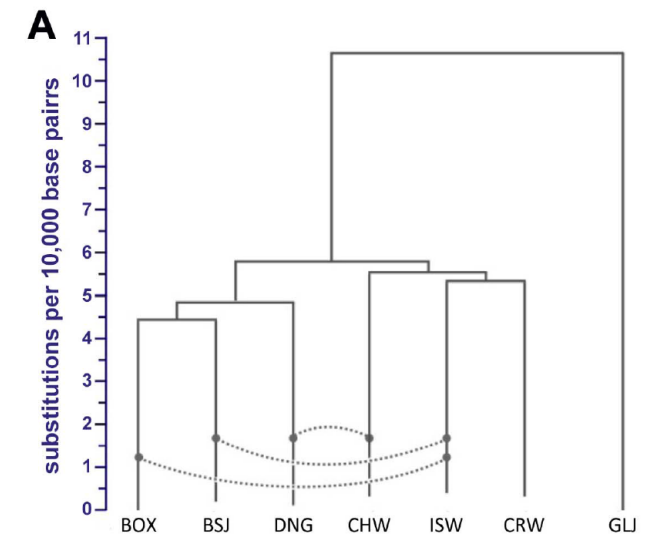
Hybridisation at different time and divergence scales

Current ongoing hybridisation:

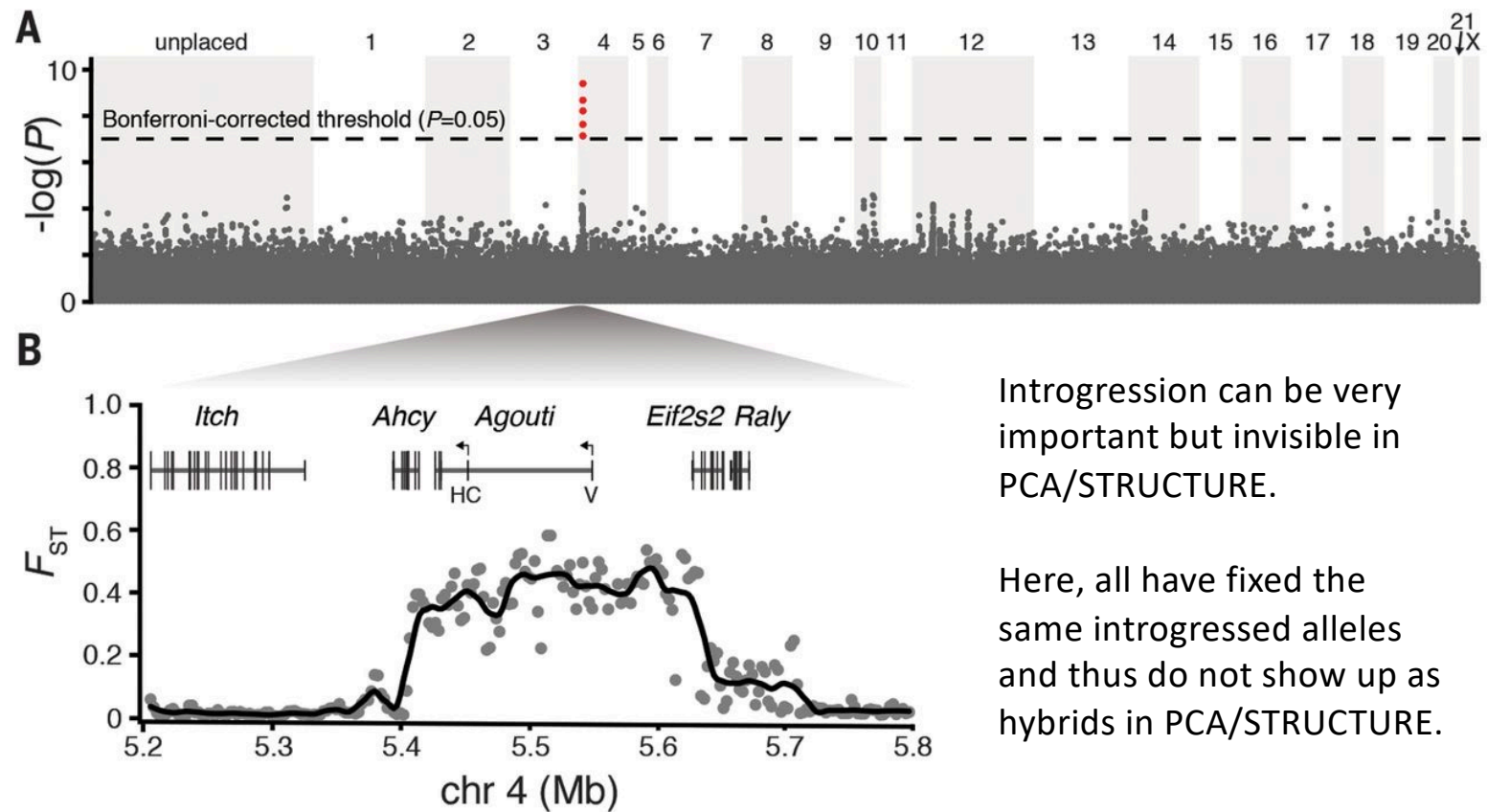
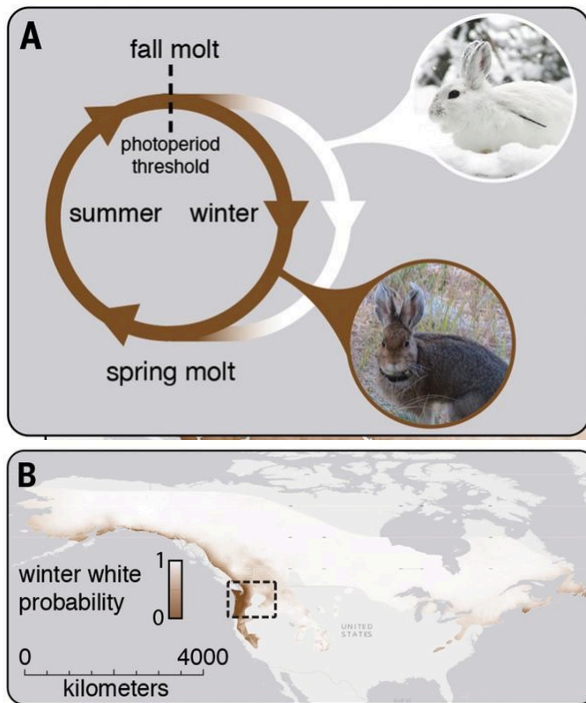
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- **Triangle plots** to classify hybrids
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Past hybridisation (introgression, potentially between distant relatives)

- Gene flow between sister species / closely related groups:
 - Demographic modelling (e.g. fastsimcoal, GPhoCS)
 - FST/Dxy
 - fd scans if unadmixed sister population/species is available
- Between more distant relatives:
 - D statistics (=ABBA/BABA test), F branch
 - Phylogenetic methods that include introgression (e.g. PhyloNet, SpeciesNetwork, AIM, treemix, etc)
 - Comparing phylogenies at different loci (e.g. TWISST)



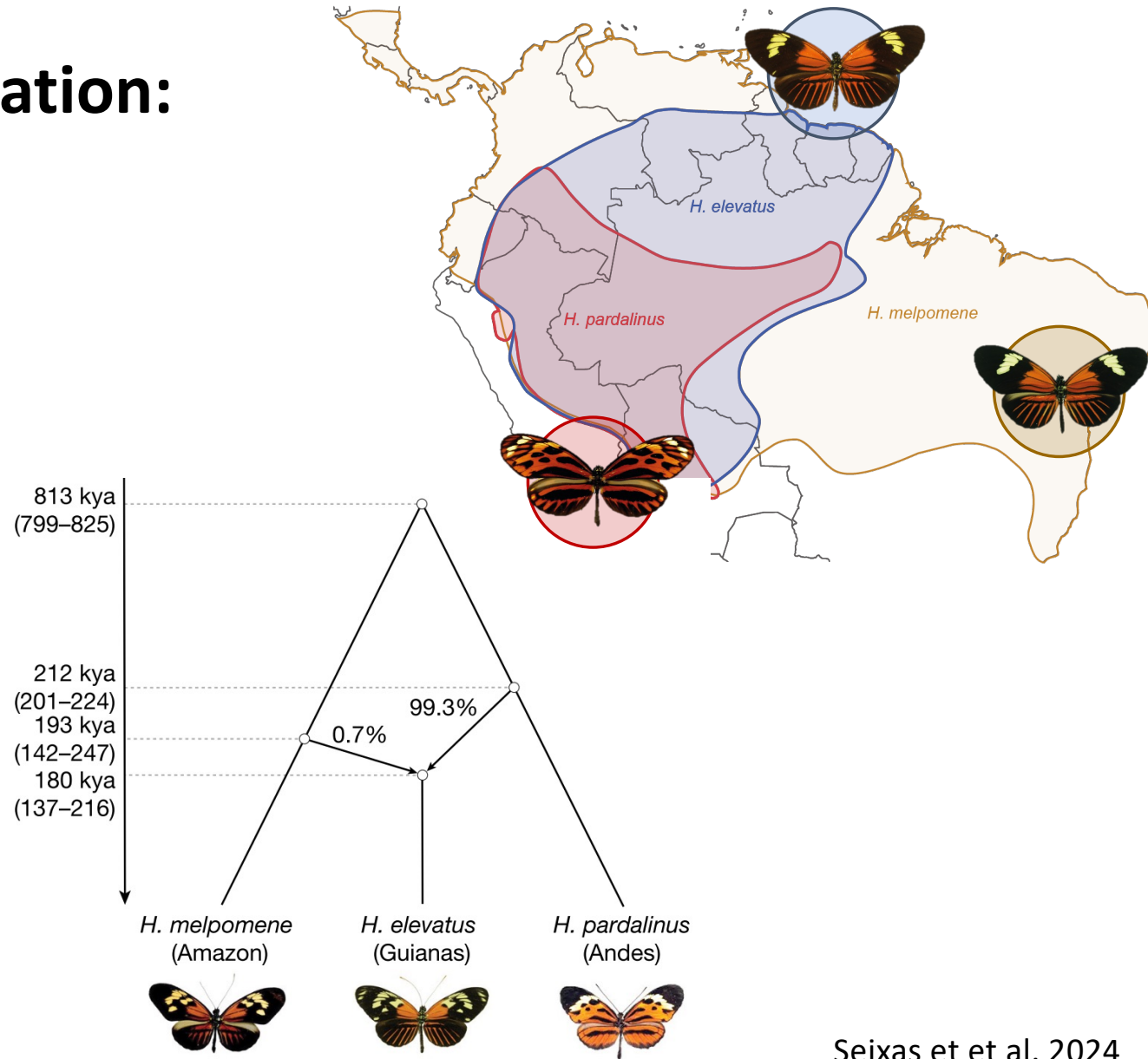
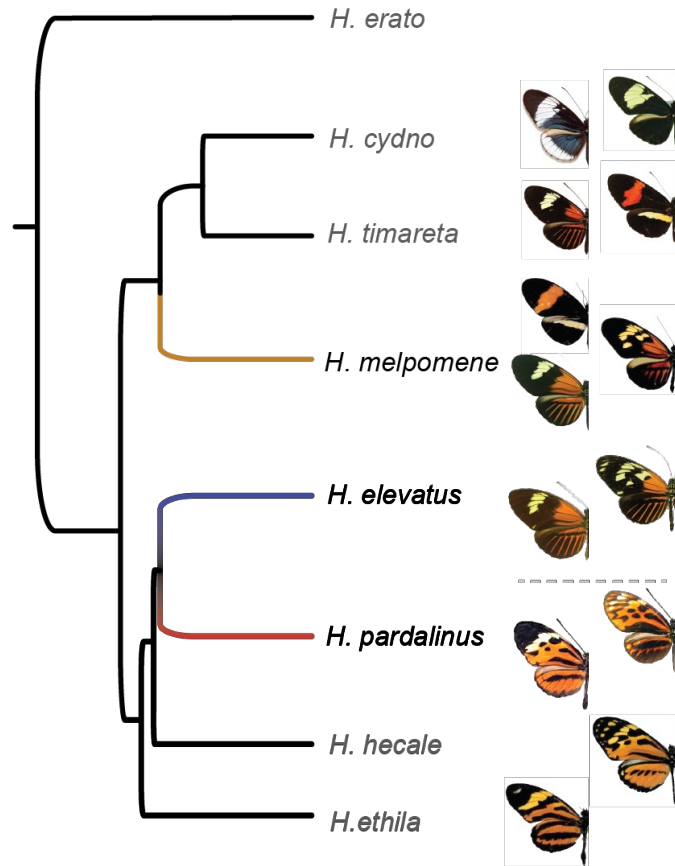
Example of adaptation to climate change: Adaptive introgression in snowshoe hares



Introgression can be very important but invisible in PCA/STRUCTURE.

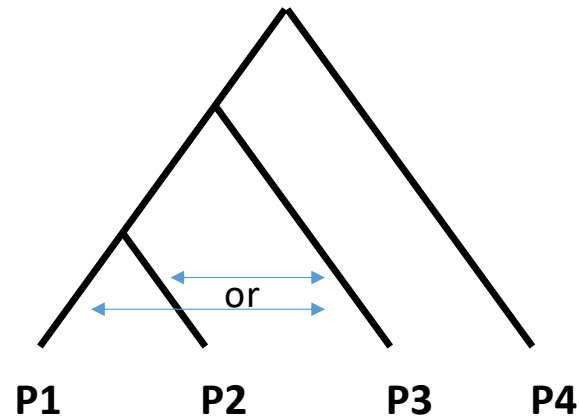
Here, all have fixed the same introgressed alleles and thus do not show up as hybrids in PCA/STRUCTURE.

Example of hybrid speciation:



Seixas et al. 2024

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

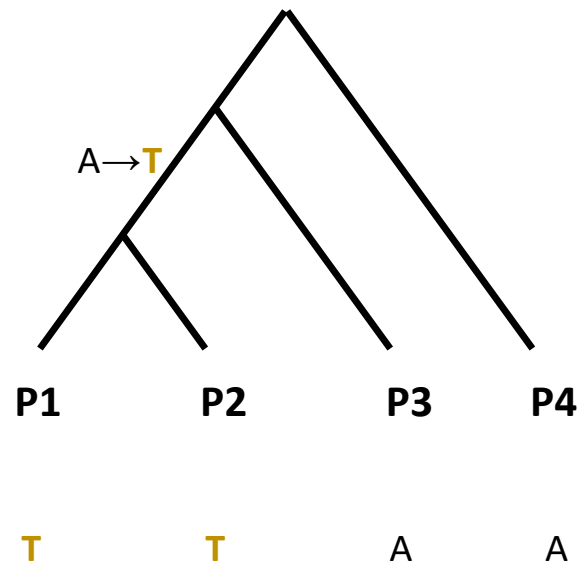
P1: T---G---C---C---A--

P2: T---G---T---A---T--

P3: A---G---T---C---T--

P4: A---T---T---A---A-- (outgroup)

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

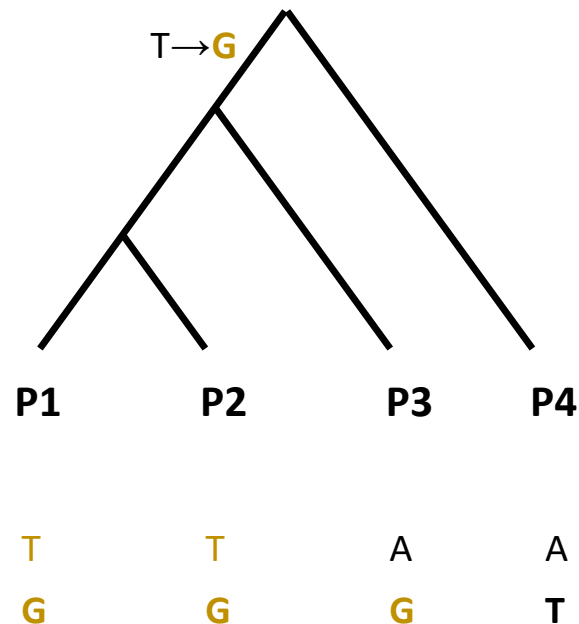
P1: T---G---C---C---A--

P2: T---G---T---A---T--

P3: A---G---T---C---T--

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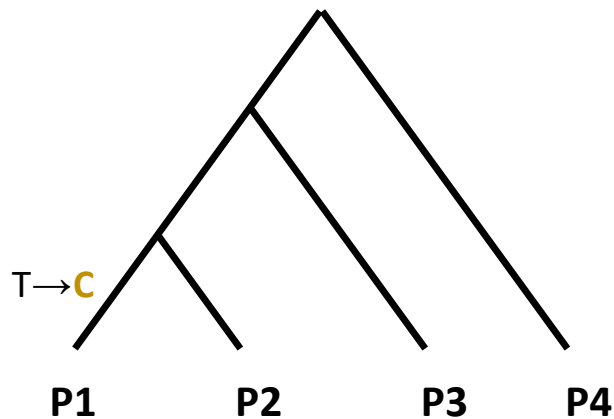
P1: T---G---C---C---A--

P2: T---G---T---A---T--

P3: A---G---T---C---T--

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P1: T---G---C---C---A--

P2: T---G---T---A---T--

P3: A---G---T---C---T--

P4: A---T---T---A---A-- (outgroup)

T
G
C

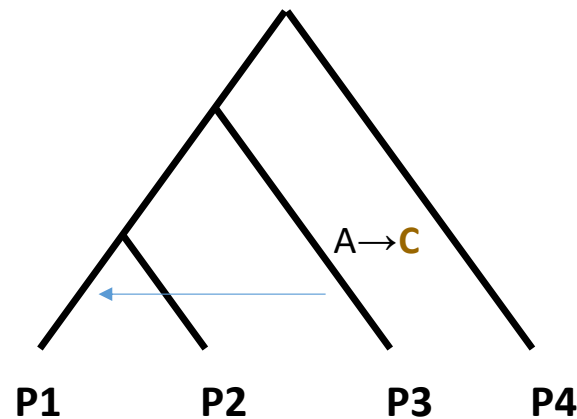
T
G
T

A
G
T

A
T
T

Concordant SNPs

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---**C**---A--

P2: T---G---T---**A**---T--

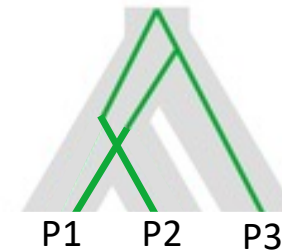
P3: A---G---T---**C**---T--

P4: A---T---T---**A**---A-- (outgroup)

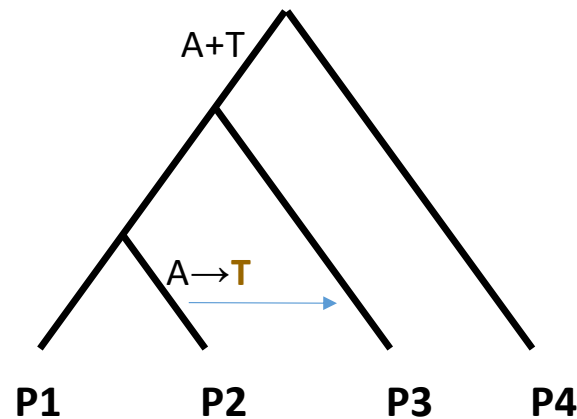
T	T	A	A
G	G	G	T
C	T	T	T
C	A	C	A

Concordant SNPs

Could also be due to incomplete Lineage sorting



Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---C---A--

P2: T---G---T---A---T--

P3: A---G---T---C---T--

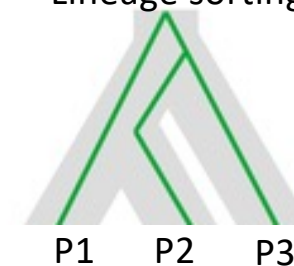
P4: A---T---T---A---A-- (outgroup)

T	T	A	A
G	G	G	T
C	T	T	T
C	A	C	A
A	T	T	A

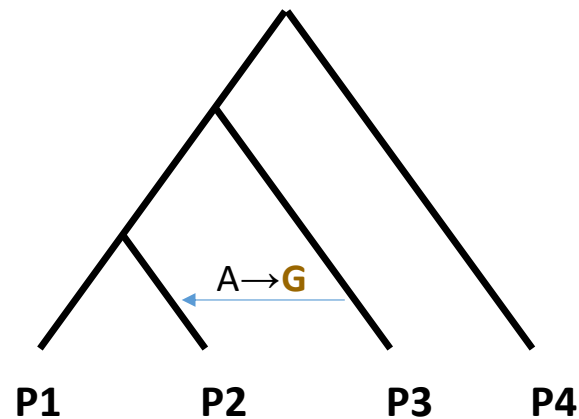
Concordant SNPs

Discordant SNPs

Could also be due to incomplete Lineage sorting



Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---C---A--A-

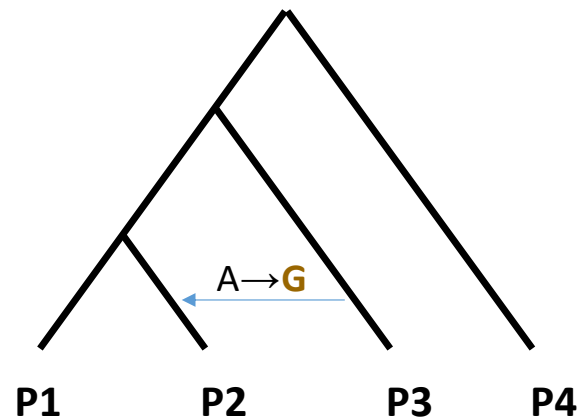
P2: T---G---T---A---T--**G**-

P3: A---G---T---C---T--**G**-

P4: A---T---T---A---A--A- (outgroup)

T	T	A	A	} Concordant SNPs
G	G	G	T	
C	T	T	T	
C	A	C	A	} Discordant SNPs
A	T	T	A	
A	G	G	A	

Patterson's D statistics (or ABBA-BABA test) to identify hybridisation between non-sister lineages



Testing for gene flow between P3 and either P1 or P2

Let's imagine these are our sequences

P1: T---G---C---C---A--A-

P2: T---G---T---A---T--G-

P3: A---G---T---C---T--G-

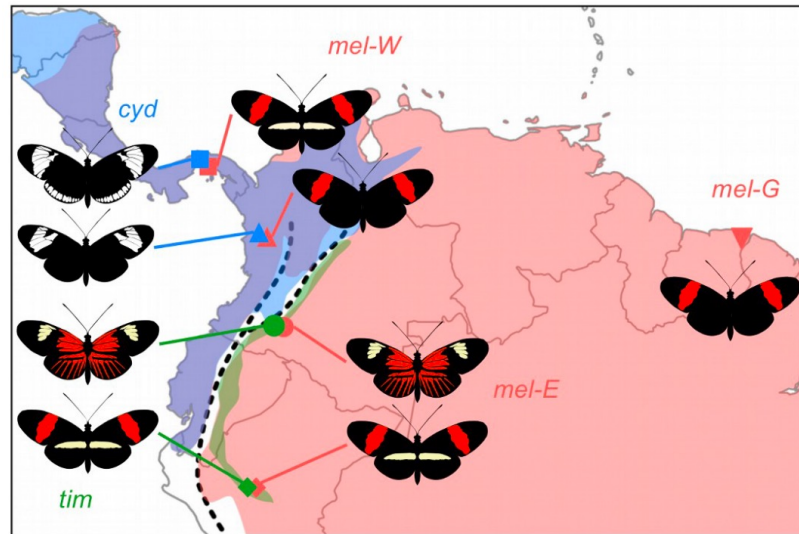
P4: A---T---T---A---A--A- (outgroup)

T	T	A	A	} Concordant SNPs
G	G	G	T	
C	T	T	T	
C	A	C	A	
A	T	T	A	← BABA
A	G	G	A	← ABBA
				← ABBA

$$D = \frac{ABBA - BABA}{ABBA + BABA}$$

$$D = \frac{2-1}{2+1} = 1/3$$

We will now do a tutorial to compute D statistics with these *Heliconius* butterflies with Dsuite



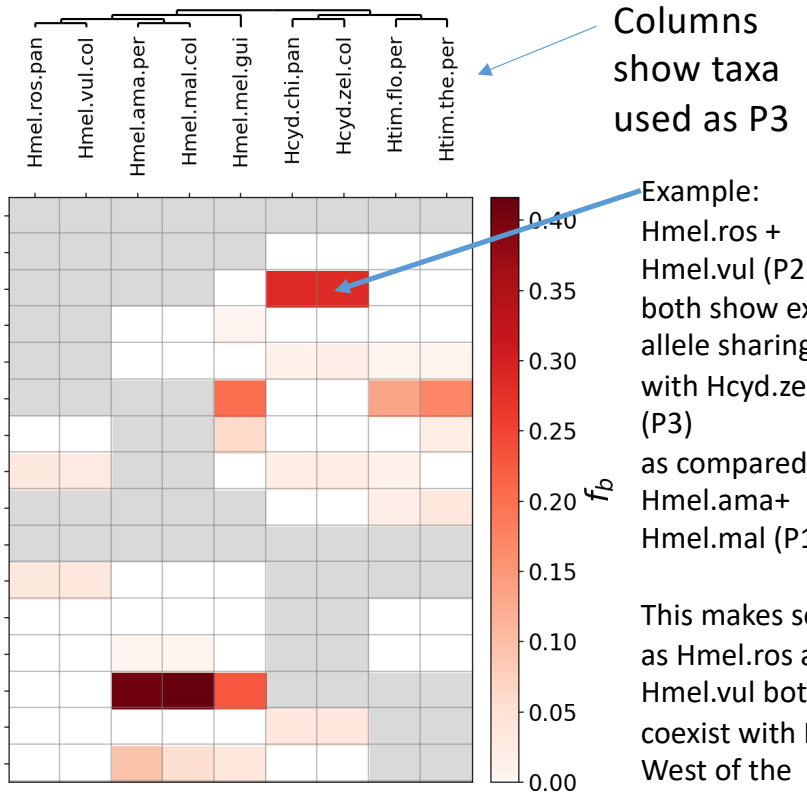
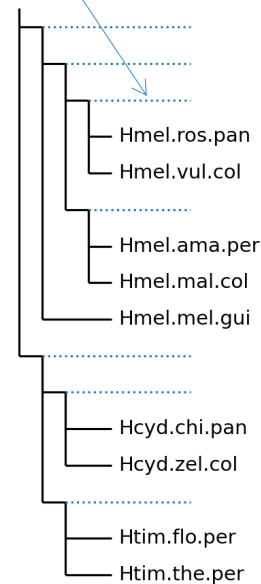
- *H. melpomene rosina*
- ▲ *H. m. vulcanus*
- *H. m. malleti*
- ◆ *H. m. amaryllis*
- ▼ *H. m. melpomene*
- *H. cydno chioneus*
- ▲ *H. c. zeline*
- *H. timareta florenxia*
- ◆ *H. t. thelxinoe*

D statistics can be summarised in Dsuite in **Fbranch**

Below an example for these butterflies

(each line shows excess allele sharing with the taxon above compared to the closest relative)

These lines refer to the group including both Hmel.ros and Hmel.vul



Example:
Hmel.ros +
Hmel.vul (P2)
both show excess
allele sharing
with Hcyd.zel
(P3)
as compared to
Hmel.ama+
Hmel.mal (P1)

This makes sense
as Hmel.ros and
Hmel.vul both
coexist with Hcyd
West of the
Andes

Supplementary

Model assumptions of ADMIXTURE/STRUCTURE

- SNPs are unlinked -> LD-pruning is very important
- Individuals are unrelated -> Do not include siblings
- Populations are represented by similar number of individuals -> strongly underrepresented populations may not be detected
- Ideally use at least 10,000 SNPs
- Sites are bi-allelic and singletons are removed

Limitations of ADMIXTURE and similar methods

ADMIXTURE and similar methods are great at inferring how many groups of individuals the dataset contains and which ones belong to which group. However, they struggle with the following:

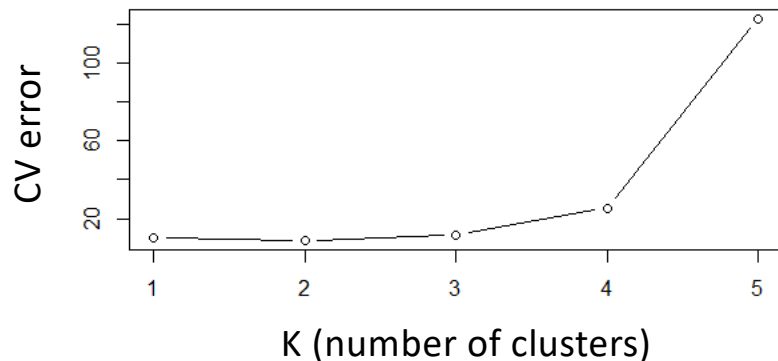
- Detection of clusters that are strongly undersampled (e.g. just a single very divergent individual may not be picked up)
- Detecting ancestral admixture that is shared by all individuals of a cluster
- Handling individuals that are highly inbred or with loads of missing data

How to identify the best number of clusters?

ADMIXTURE uses a cross-validation (CV) procedure:

e.g. 5-fold CV: the genotypes are randomly split into five partitions. Each partition in turn is masked (set to missing) and then predicted with the individual cluster assignment and allele frequencies inferred for each cluster. The predicted genotypes are compared to the real genotypes to infer the prediction error. The prediction errors together are used to infer the cross validation error.

The lowest CV error indicates the best number of K but don't just use the CV error.



Make sure to always look at the plots with different numbers of clusters and also use your knowledge on the biology and collection places to identify which number of clusters is most informative.

STRUCTURE vs ADMIXTURE

- STRUCTURE is a model-based clustering approach which utilizes genotype data to infer the presence of distinct populations and assign individuals to populations. The number of populations (k) has to be predefined by the user. Usually, one tests multiple values and checks at which k the model posterior probability start to level out. Using an MCMC approach, STRUCTURE identifies k clusters that are each in HWE and assigns individuals to these clusters or a mix of them.
- ADMIXTURE works similarly but uses a maximum likelihood estimation to identify clusters and infer individual ancestries. The best value of k is determined with cross-validation.
- ADMIXTURE is much faster than STRUCTURE and thus ideal for large NGS datasets. STRUCTURE may be better suited for identification of very weak population structure also because it can use prior probabilities for the cluster assignments. However, ADMIXTURE allows to specification of unadmixed reference individuals with known ancestry.
- There are other similar tools that are faster, like fastSTRUCTURE or PopCluster
See a comparison of these tools here: <https://onlinelibrary.wiley.com/doi/10.1111/1755-0998.14058>

Is the ADMIXTURE plot a good fit for your data?

- **badmixture**
(Dan Lawson)
- **Evaladmix**
(Garcia-Erill & Albrechtsen)

Positive correlation of residuals (red) indicates that the individuals in this group are more closely related than ADMIXTURE suggests, blue means that they are less closely related than modelled by ADMIXTURE.

